Step 5 Fixed-Window Onset-Locked ROI Follow-Up

This report documents the targeted onset-locked follow-up requested after the non-significant pre-specified Step 4 primary test. Step 4 remains the primary analysis; Step 5 is a focused follow-up using the same participant-session cohort and a fixed 7–11 s post-onset HbO window.

Overview

Metric	Value
Step 4 included participant-sessions reused	36
Step 5 included participant-sessions	36
Valid onset-locked trials in Step 5	1075
Primary valid abstract trials	539
Primary valid concrete trials	536
Primary window	7.0 s to 11.0 s after onset
Sensitivity windows	6.0-10.0 s, 8.0-12.0 s, and single time point at 9.0 s
Fast-response sensitivity	Exclude trials with answer latency ¡ 11.0 s

Shared Global Exclusions

The shared exclusion list remained active in Step 5 and the excluded participants were not reintroduced.

Metric	Value
PID006	Global downstream exclusion after Step 3 review: Excluded from Step 1 and Step 2 summaries and from future downstream analyses after Step 3 documented session 2025-11-12_002 as an implausibly short 10.420224 second recording with no paired PsychoPy log.
PID037	Global downstream exclusion after Step 3 review: Excluded from Step 1 and Step 2 summaries and from future downstream analyses because a behavioral log exists but no fNIRS session folder was found in Step 3.

Inherited Step 4 Manual Inclusions

Step 5 reused the Step 4 analysis cohort, including the following explicitly approved manual inclusions.

Metric	Value
PID025 / 2025-11-25_002	User-directed Step 4 override added on 2026-03-20: include this session despite Step 3 REVIEW_BEFORE_STEP4 status.
PID034 / 2025-11-27_003	User-directed Step 4 override added on 2026-03-20: include this session despite Step 3 REVIEW_BEFORE_STEP4 status.

Primary Front ROI Fixed-Window Test

n	Abstract	Concrete	Mean diff	t	df	p	95% CI	d_z
36	0.0000	-0.0000	0.0000	1.8930	35	0.066656	[-0.0000, 0.0000]	0.3155

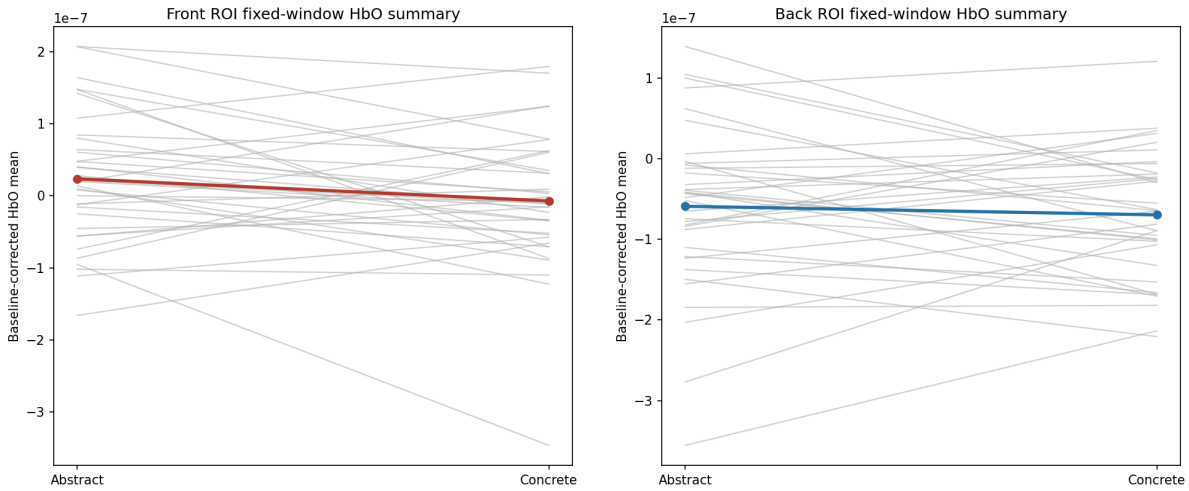

Figure 1: Participant-level fixed-window HbO summaries for the front and back ROIs.

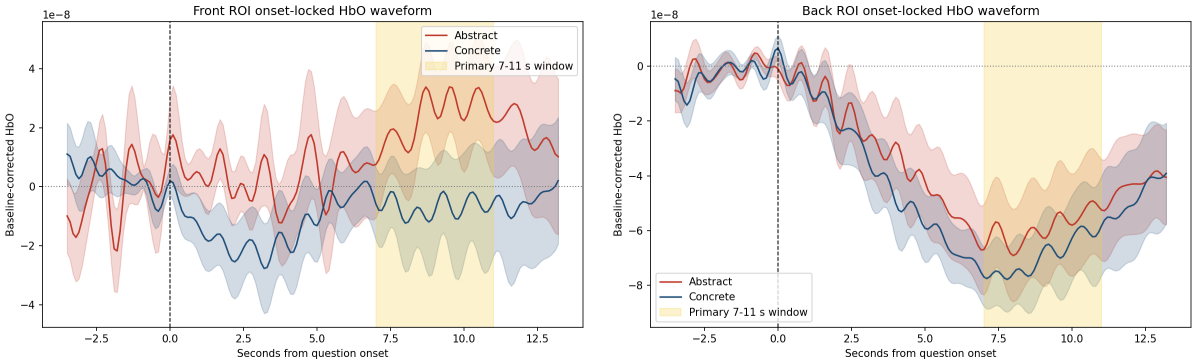

Figure 2: Descriptive onset-locked HbO waveforms with the 7–11 s primary analysis window shaded.

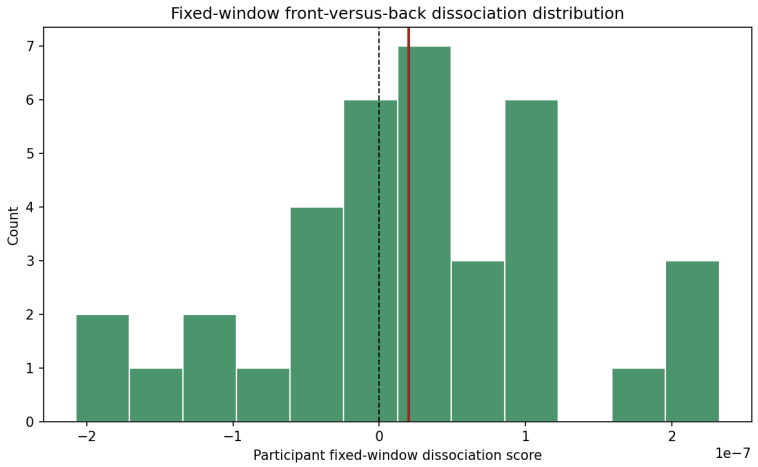

Figure 3: Distribution of the participant-level fixed-window HbO dissociation scores.

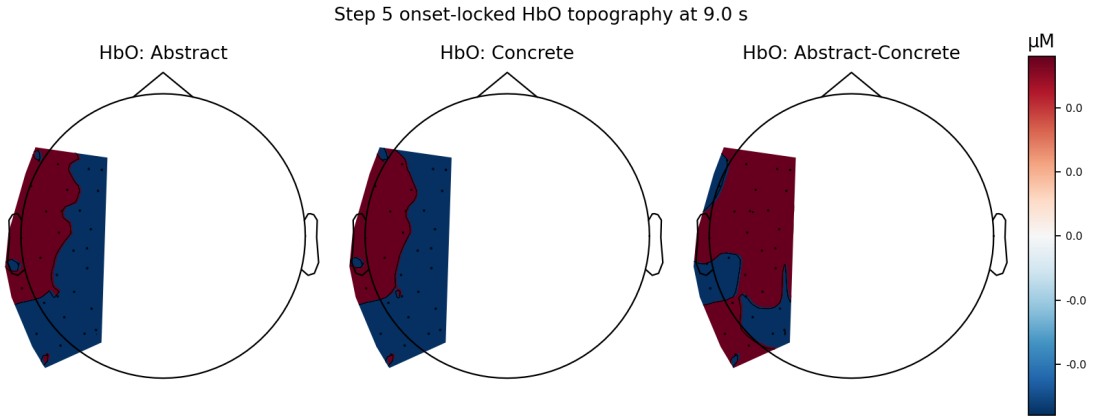

Figure 4: HbO topographic comparison at 9.0 s after question onset for Abstract, Concrete, and Abstract-minus-Concrete.

Secondary and Sensitivity Results

Analysis	n	Mean	t	df	p	95% CI	d_z
HBO Back window_7_11	36	0.0000	0.7123	35	0.480986	[-0.0000, 0.0000]	0.1187
HBR Back window_7_11	36	-0.0000	-0.3675	35	0.715475	[-0.0000, 0.0000]	-0.0612
HBO nan window_7_11	36		1.1621	35	0.253048	[-0.0000, 0.0000]	0.1937
HBR nan window_7_11	36		0.4402	35	0.662481	[-0.0000, 0.0000]	0.0734
HBO Front timepoint_9s	36	0.0000	1.7890	35	0.082266	[-0.0000, 0.0000]	0.2982
HBO Back timepoint_9s	36	0.0000	0.3160	35	0.753893	[-0.0000, 0.0000]	0.0527
HBO nan timepoint_9s	36		1.5180	35	0.138000	[-0.0000, 0.0000]	0.2530
HBR Front timepoint_9s	36	0.0000	0.3112	35	0.757508	[-0.0000, 0.0000]	0.0519
HBR Back timepoint_9s	36	-0.0000	-0.4150	35	0.680658	[-0.0000, 0.0000]	-0.0692
HBR nan timepoint_9s	36		0.5263	35	0.601994	[-0.0000, 0.0000]	0.0877
HBO Front window_6_10	36	0.0000	1.7777	35	0.084140	[-0.0000, 0.0000]	0.2963
HBO Back window_6_10	36	0.0000	0.8075	35	0.424840	[-0.0000, 0.0000]	0.1346
HBO nan window_6_10	36		0.7735	35	0.444397	[-0.0000, 0.0000]	0.1289
HBR Front window_6_10	36	-0.0000	-0.2823	35	0.779411	[-0.0000, 0.0000]	-0.0470
HBR Back window_6_10	36	-0.0000	-0.2915	35	0.772360	[-0.0000, 0.0000]	-0.0486
HBR nan window_6_10	36		-0.0786	35	0.937768	[-0.0000, 0.0000]	-0.0131
HBO Front window_8_12	36	0.0000	1.8261	35	0.076372	[-0.0000, 0.0000]	0.3044
HBO Back window_8_12	36	0.0000	0.5534	35	0.583505	[-0.0000, 0.0000]	0.0922
HBO nan window_8_12	36		1.3190	35	0.195735	[-0.0000, 0.0000]	0.2198
HBR Front window_8_12	36	0.0000	0.6597	35	0.513773	[-0.0000, 0.0000]	0.1099
HBR Back window_8_12	36	-0.0000	-0.4249	35	0.673495	[-0.0000, 0.0000]	-0.0708
HBR nan window_8_12	36		0.8658	35	0.392490	[-0.0000, 0.0000]	0.1443
HBO Front window_7_11_fast_response	36	0.0000	1.8930	35	0.066656	[-0.0000, 0.0000]	0.3155

Primary follow-up result. Front HbO p=0.066656; back HbO p=0.480986; dissociation HbO p=0.253048.

Final Conclusion

In the targeted fixed-window onset-locked follow-up analysis, the difference between abstract and concrete questions in the left frontal ROI during the 7-11 s post-onset window was not statistically significant (t(35)=1.8930, p=0.066656, mean difference=0.0000, 95